```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.datasets import imdb
import random
```

```
# Set seed for reproducibility
def initialize_seed(seed_value=42):
    """
    Initializes the random seed for NumPy, TensorFlow, and Python random.
    """
    np.random.seed(seed_value)
    tf.random.set_seed(seed_value)
    random.seed(seed_value)
```

```
initialize_seed(42)
```

```
# Load IMDB dataset
print("Loading IMDB dataset: ")
(train_reviews, train_labels), (test_reviews, test_labels) = imdb.load_data(num_words=10000)
```

```
Loading IMDB dataset:
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
# Function to convert sequences into binary multi-hot encoding
def encode_sequences(sequences, vocab_size=10000):
    """
    Converts sequences of word indices into a binary multi-hot encoded matrix.
    """
    output_matrix = np.zeros((len(sequences), vocab_size))
    for idx, seq in enumerate(sequences):
        output_matrix[idx, seq] = 1.0
    return output_matrix
```

```
# Transform training and testing data into vectorized format
x_train = encode_sequences(train_reviews)
x_test = encode_sequences(test_reviews)
```

```
y_train = np.asarray(train_labels).astype("float32")
y_test = np.asarray(test_labels).astype("float32")
```

```
# Split the data into training and validation sets
x_validation = x_train[:10000]
x_partial_train = x_train[10000:]
y_validation = y_train[:10000]
y_partial_train = y_train[10000:]
```

```
# Function to construct the neural network model
def construct_model(hidden_layers=2, neurons=16, activation_fn='relu', dropout=0.0, loss_fn='binary_crossentropy'):
    """
    Builds and compiles a sequential neural network model.
    """
    model = tf.keras.Sequential()
    model.add(layers.Dense(neurons, activation=activation_fn, input_shape=(10000,)))

    for _ in range(hidden_layers):
        model.add(layers.Dense(neurons, activation=activation_fn))
    if dropout > 0.0:
        model.add(layers.Dropout(dropout))
    model.add(layers.Dense(1, activation='sigmoid'))
    model.compile(optimizer='rmsprop',
```

```
                    loss=loss_fn,
                    metrics=['accuracy', 'precision', 'recall', 'auc'])
    return model
```

```
# Dictionary to store experiment results
experiment_results = {
'Hidden Layers': {},
'Neuron Count': {},
'Loss Function': {},
'Activation Function': {},
'Dropout Rate': {}
}
```

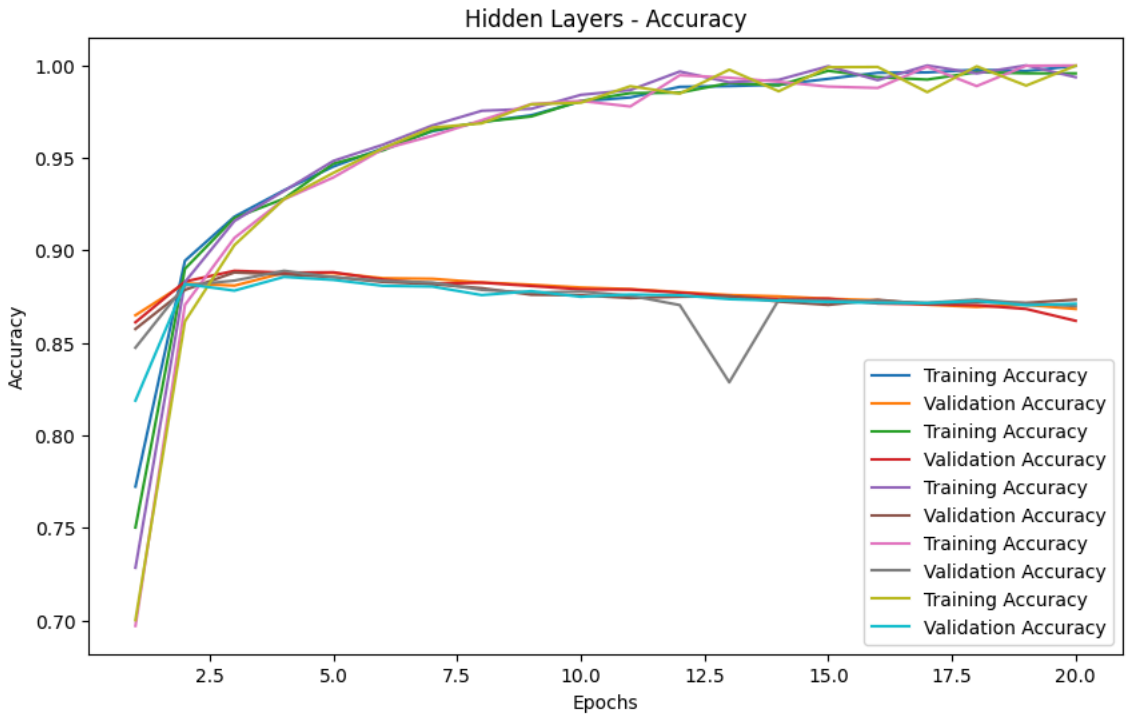
```
# Function to visualize training and validation metrics
def visualize_metrics(history_dict, title, y_label):
    """
    Plots accuracy or loss over epochs for different experimental setups.
    """
    plt.figure(figsize=(10, 6))
    for label, history in history_dict.items():
        epochs = range(1, len(history.history['accuracy']) + 1)
        plt.plot(epochs, history.history['accuracy'], label=f'Training {y_label}')
        plt.plot(epochs, history.history['val_accuracy'], label=f'Validation {y_label}')
    plt.title(title)
    plt.xlabel('Epochs')
    plt.ylabel(y_label)
    plt.legend()
    plt.show()
```

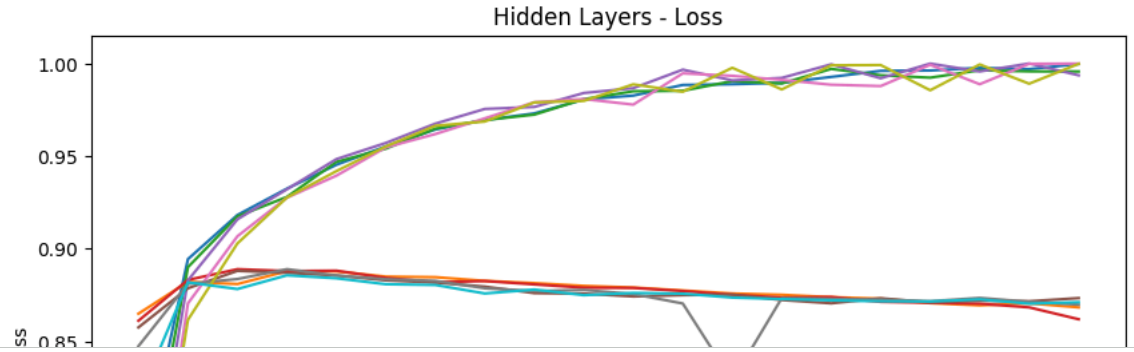
```
# Experiment 1: Testing different numbers of hidden layers
print("\n--- Experiment 1: Number of Hidden Layers ---")
hidden_layer_configs = [1, 2, 3, 4, 5]
histories_hidden_layers = {}

for num_layers in hidden_layer_configs:
    print(f"Training model with {num_layers} hidden layer(s)...")
    model = construct_model(hidden_layers=num_layers, neurons=16)
    history = model.fit(x_partial_train, y_partial_train,
                        epochs=20,
                        batch_size=512,
                        validation_data=(x_validation, y_validation),
                        verbose=1)
    histories_hidden_layers[num_layers] = history
    val_acc = history.history['val_accuracy'][-1]
    test_acc = model.evaluate(x_test, y_test, verbose=0)[1]
    experiment_results['Hidden Layers'][num_layers] = {'Validation Accuracy': val_acc, 'Test Accuracy': test_acc}


visualize_metrics(histories_hidden_layers, 'Hidden Layers - Accuracy', 'Accuracy')
visualize_metrics(histories_hidden_layers, 'Hidden Layers - Loss', 'Loss')
```

```
--- Experiment 1: Number of Hidden Layers ---
Training model with 1 hidden layer(s)...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 134ms/step - accuracy: 0.6929 - auc: 0.7612 - loss: 0.6123 - precision: 0.6749 - recall: 0.7535 -
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 45ms/step - accuracy: 0.8907 - auc: 0.9519 - loss: 0.3598 - precision: 0.8804 - recall: 0.9029 - v
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 37ms/step - accuracy: 0.9155 - auc: 0.9717 - loss: 0.2579 - precision: 0.9086 - recall: 0.9228 - v
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - accuracy: 0.9311 - auc: 0.9792 - loss: 0.2091 - precision: 0.9305 - recall: 0.9311 - v
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9462 - auc: 0.9861 - loss: 0.1703 - precision: 0.9448 - recall: 0.9470 - v
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.9577 - auc: 0.9903 - loss: 0.1417 - precision: 0.9583 - recall: 0.9565 - v
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9659 - auc: 0.9931 - loss: 0.1202 - precision: 0.9679 - recall: 0.9634 - v
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9715 - auc: 0.9950 - loss: 0.1026 - precision: 0.9728 - recall: 0.9699 - v
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 44ms/step - accuracy: 0.9759 - auc: 0.9962 - loss: 0.0894 - precision: 0.9769 - recall: 0.9745 - v
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - accuracy: 0.9819 - auc: 0.9976 - loss: 0.0744 - precision: 0.9831 - recall: 0.9805 - v
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9843 - auc: 0.9981 - loss: 0.0654 - precision: 0.9840 - recall: 0.9843 - v
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9898 - auc: 0.9988 - loss: 0.0537 - precision: 0.9902 - recall: 0.9893 - v
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9899 - auc: 0.9991 - loss: 0.0494 - precision: 0.9897 - recall: 0.9900 - v
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9892 - auc: 0.9991 - loss: 0.0458 - precision: 0.9902 - recall: 0.9881 - v
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9946 - auc: 0.9996 - loss: 0.0322 - precision: 0.9948 - recall: 0.9944 - v
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9961 - auc: 0.9998 - loss: 0.0278 - precision: 0.9965 - recall: 0.9957 - v
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9967 - auc: 0.9998 - loss: 0.0259 - precision: 0.9967 - recall: 0.9967 - v
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 31ms/step - accuracy: 0.9983 - auc: 1.0000 - loss: 0.0204 - precision: 0.9986 - recall: 0.9980 - v
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9985 - auc: 1.0000 - loss: 0.0160 - precision: 0.9988 - recall: 0.9982 - v
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9998 - auc: 1.0000 - loss: 0.0119 - precision: 0.9998 - recall: 0.9998 - v
Training model with 2 hidden layer(s)...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 71ms/step - accuracy: 0.6597 - auc: 0.7249 - loss: 0.6306 - precision: 0.6543 - recall: 0.6716 - v
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.8844 - auc: 0.9468 - loss: 0.3619 - precision: 0.8818 - recall: 0.8863 - v
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9151 - auc: 0.9699 - loss: 0.2553 - precision: 0.9137 - recall: 0.9157 - v
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9275 - auc: 0.9785 - loss: 0.2049 - precision: 0.9261 - recall: 0.9281 - v
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9499 - auc: 0.9872 - loss: 0.1603 - precision: 0.9496 - recall: 0.9496 - v
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.9577 - auc: 0.9909 - loss: 0.1343 - precision: 0.9589 - recall: 0.9559 - v
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 35ms/step - accuracy: 0.9672 - auc: 0.9938 - loss: 0.1117 - precision: 0.9671 - recall: 0.9669 - v
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9701 - auc: 0.9954 - loss: 0.0968 - precision: 0.9713 - recall: 0.9685 - v
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - accuracy: 0.9747 - auc: 0.9965 - loss: 0.0848 - precision: 0.9753 - recall: 0.9738 - v
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - accuracy: 0.9839 - auc: 0.9981 - loss: 0.0652 - precision: 0.9844 - recall: 0.9831 - v
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9873 - auc: 0.9987 - loss: 0.0551 - precision: 0.9878 - recall: 0.9866 - v
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9897 - auc: 0.9991 - loss: 0.0472 - precision: 0.9894 - recall: 0.9898 - v
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9930 - auc: 0.9995 - loss: 0.0372 - precision: 0.9938 - recall: 0.9920 - v
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9931 - auc: 0.9996 - loss: 0.0321 - precision: 0.9933 - recall: 0.9929 - v
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 45ms/step - accuracy: 0.9971 - auc: 0.9998 - loss: 0.0233 - precision: 0.9971 - recall: 0.9971 - v
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9893 - auc: 0.9984 - loss: 0.0390 - precision: 0.9880 - recall: 0.9906 - v
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9872 - auc: 0.9983 - loss: 0.0405 - precision: 0.9848 - recall: 0.9895 - v
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9981 - auc: 0.9999 - loss: 0.0143 - precision: 0.9984 - recall: 0.9978 - v
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9989 - auc: 1.0000 - loss: 0.0114 - precision: 0.9986 - recall: 0.9991 - v
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9993 - auc: 1.0000 - loss: 0.0088 - precision: 0.9992 - recall: 0.9993 - v
```

```
Training model with 3 hidden layer(s)...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 85ms/step - accuracy: 0.6385 - auc: 0.7156 - loss: 0.6490 - precision: 0.6895 - recall: 0.4991 - v
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.8804 - auc: 0.9431 - loss: 0.3811 - precision: 0.8786 - recall: 0.8813 - v
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9164 - auc: 0.9705 - loss: 0.2491 - precision: 0.9147 - recall: 0.9174 - v
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9324 - auc: 0.9802 - loss: 0.1929 - precision: 0.9293 - recall: 0.9352 - v
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9531 - auc: 0.9880 - loss: 0.1496 - precision: 0.9522 - recall: 0.9535 - v
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9604 - auc: 0.9921 - loss: 0.1206 - precision: 0.9591 - recall: 0.9614 - v
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9708 - auc: 0.9947 - loss: 0.0972 - precision: 0.9693 - recall: 0.9720 - v
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9776 - auc: 0.9965 - loss: 0.0780 - precision: 0.9773 - recall: 0.9777 - v
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 37ms/step - accuracy: 0.9796 - auc: 0.9972 - loss: 0.0678 - precision: 0.9794 - recall: 0.9796 - v
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9898 - auc: 0.9988 - loss: 0.0469 - precision: 0.9891 - recall: 0.9904 - v
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.9919 - auc: 0.9989 - loss: 0.0381 - precision: 0.9926 - recall: 0.9912 - v
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9964 - auc: 0.9997 - loss: 0.0229 - precision: 0.9966 - recall: 0.9962 - v
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9869 - auc: 0.9989 - loss: 0.0394 - precision: 0.9871 - recall: 0.9866 - v
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9884 - auc: 0.9989 - loss: 0.0363 - precision: 0.9902 - recall: 0.9863 - v
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9994 - auc: 1.0000 - loss: 0.0083 - precision: 0.9995 - recall: 0.9993 - v
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9844 - auc: 0.9978 - loss: 0.0456 - precision: 0.9832 - recall: 0.9856 - v
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9999 - auc: 1.0000 - loss: 0.0037 - precision: 0.9998 - recall: 0.9999 - v
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step - accuracy: 0.9932 - auc: 0.9998 - loss: 0.0178 - precision: 0.9932 - recall: 0.9930 - v
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 43ms/step - accuracy: 0.9999 - auc: 1.0000 - loss: 0.0019 - precision: 1.0000 - recall: 0.9999 - v
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 33ms/step - accuracy: 0.9968 - auc: 0.9993 - loss: 0.0127 - precision: 0.9962 - recall: 0.9973 - v
Training model with 4 hidden layer(s)...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 70ms/step - accuracy: 0.6079 - auc: 0.6619 - loss: 0.6703 - precision: 0.6265 - recall: 0.5432 - v
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.8617 - auc: 0.9265 - loss: 0.4337 - precision: 0.8549 - recall: 0.8696 - v
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 33ms/step - accuracy: 0.9047 - auc: 0.9603 - loss: 0.2828 - precision: 0.9009 - recall: 0.9083 - v
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9297 - auc: 0.9769 - loss: 0.2042 - precision: 0.9288 - recall: 0.9299 - v
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9425 - auc: 0.9842 - loss: 0.1634 - precision: 0.9409 - recall: 0.9437 - v
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9573 - auc: 0.9894 - loss: 0.1325 - precision: 0.9579 - recall: 0.9562 - v
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9657 - auc: 0.9925 - loss: 0.1074 - precision: 0.9644 - recall: 0.9667 - v
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9749 - auc: 0.9942 - loss: 0.0898 - precision: 0.9750 - recall: 0.9745 - v
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9764 - auc: 0.9949 - loss: 0.0795 - precision: 0.9771 - recall: 0.9753 - v
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9749 - auc: 0.9948 - loss: 0.0781 - precision: 0.9749 - recall: 0.9745 - v
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9867 - auc: 0.9967 - loss: 0.0516 - precision: 0.9873 - recall: 0.9859 - v
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9953 - auc: 0.9987 - loss: 0.0285 - precision: 0.9960 - recall: 0.9945 - v
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9906 - auc: 0.9988 - loss: 0.0347 - precision: 0.9910 - recall: 0.9901 - v
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9752 - auc: 0.9974 - loss: 0.0615 - precision: 0.9717 - recall: 0.9796 - v
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9750 - auc: 0.9929 - loss: 0.0895 - precision: 0.9786 - recall: 0.9710 - v
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9934 - auc: 0.9986 - loss: 0.0268 - precision: 0.9936 - recall: 0.9932 - v
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9989 - auc: 0.9998 - loss: 0.0083 - precision: 0.9988 - recall: 0.9990 - v
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9936 - auc: 0.9976 - loss: 0.0309 - precision: 0.9909 - recall: 0.9962 - v
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9996 - auc: 1.0000 - loss: 0.0039 - precision: 0.9994 - recall: 0.9998 - v
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9998 - auc: 1.0000 - loss: 0.0026 - precision: 0.9997 - recall: 0.9998 - v
Training model with 5 hidden layer(s)...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 72ms/step - accuracy: 0.6170 - auc: 0.6715 - loss: 0.6651 - precision: 0.5948 - recall: 0.7222 - v
```

```
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 37ms/step - accuracy: 0.8463 - auc: 0.9200 - loss: 0.4384 - precision: 0.8172 - recall: 0.8916 - v
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.8990 - auc: 0.9576 - loss: 0.2871 - precision: 0.8950 - recall: 0.9028 - v
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9288 - auc: 0.9766 - loss: 0.2068 - precision: 0.9303 - recall: 0.9263 - v
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9435 - auc: 0.9843 - loss: 0.1625 - precision: 0.9455 - recall: 0.9405 - v
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 55ms/step - accuracy: 0.9568 - auc: 0.9892 - loss: 0.1296 - precision: 0.9602 - recall: 0.9525 - v
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 35ms/step - accuracy: 0.9685 - auc: 0.9924 - loss: 0.1034 - precision: 0.9718 - recall: 0.9646 - v
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9678 - auc: 0.9927 - loss: 0.0968 - precision: 0.9690 - recall: 0.9662 - v
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9823 - auc: 0.9959 - loss: 0.0653 - precision: 0.9868 - recall: 0.9774 - v
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - accuracy: 0.9887 - auc: 0.9963 - loss: 0.0514 - precision: 0.9904 - recall: 0.9868 - v
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9920 - auc: 0.9976 - loss: 0.0379 - precision: 0.9943 - recall: 0.9896 - v
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9927 - auc: 0.9975 - loss: 0.0352 - precision: 0.9944 - recall: 0.9910 - v
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9976 - auc: 0.9987 - loss: 0.0198 - precision: 0.9991 - recall: 0.9961 - v
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9964 - auc: 0.9984 - loss: 0.0218 - precision: 0.9987 - recall: 0.9940 - v
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 37ms/step - accuracy: 0.9986 - auc: 0.9992 - loss: 0.0123 - precision: 0.9998 - recall: 0.9973 - v
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.9987 - auc: 0.9995 - loss: 0.0098 - precision: 0.9998 - recall: 0.9975 - v
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9748 - auc: 0.9893 - loss: 0.1116 - precision: 0.9830 - recall: 0.9660 - v
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9991 - auc: 0.9997 - loss: 0.0058 - precision: 0.9998 - recall: 0.9984 - v
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9775 - auc: 0.9938 - loss: 0.0843 - precision: 0.9684 - recall: 0.9875 - v
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9997 - auc: 1.0000 - loss: 0.0029 - precision: 0.9998 - recall: 0.9996 - v
```

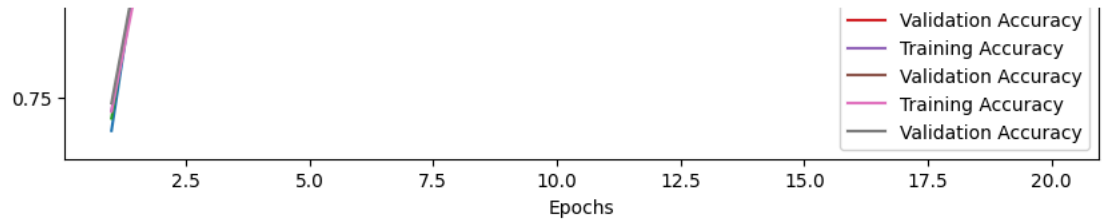
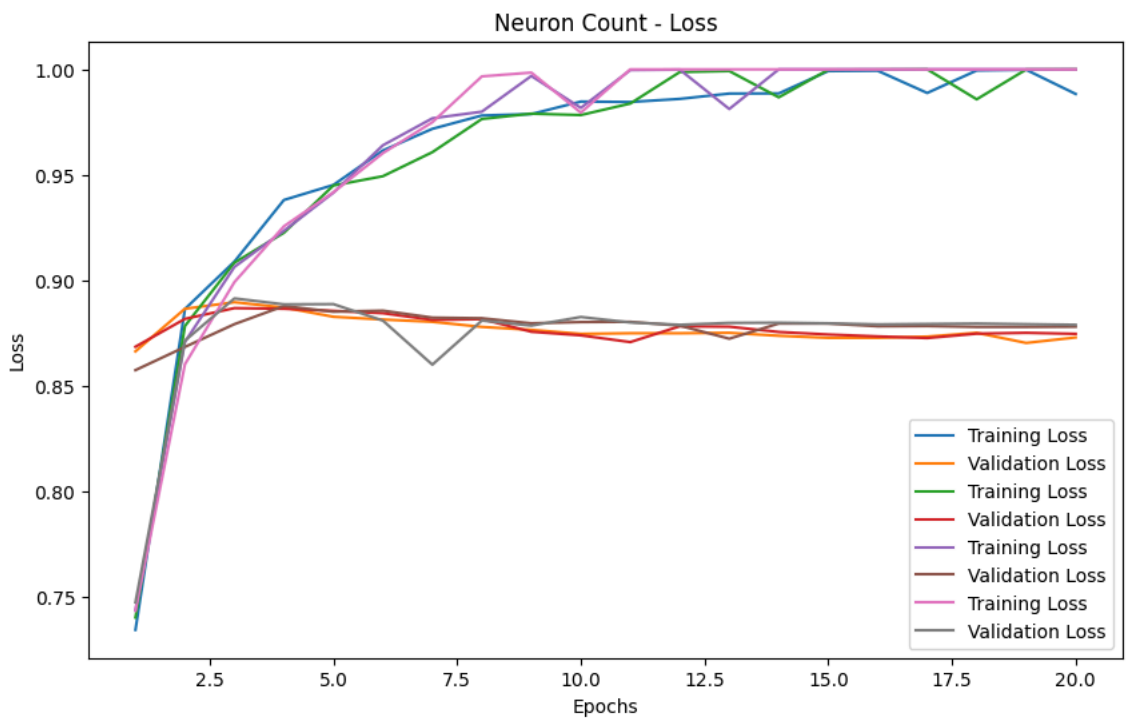




```
# Experiment 2: Changing number of neurons
print("\n--- Experiment 2: Varying Neuron Count ---")
neuron_options = [32, 64, 128, 256]
histories_neurons = {}
for neurons in neuron_options:
    print(f"Training model with {neurons} neurons per layer...")
    model = construct_model(hidden_layers=2, neurons=neurons)
    history = model.fit(x_partial_train, y_partial_train,
                        epochs=20,
                        batch_size=512,
                        validation_data=(x_validation, y_validation),
                        verbose=1)
    histories_neurons[neurons] = history
    val_acc = history.history['val_accuracy'][-1]
    test_acc = model.evaluate(x_test, y_test, verbose=0)[1]
    experiment_results['Neuron Count'][neurons] = {'Validation Accuracy': val_acc, 'Test Accuracy': test_acc}

visualize_metrics(histories_neurons, 'Neuron Count - Accuracy', 'Accuracy')
visualize_metrics(histories_neurons, 'Neuron Count - Loss', 'Loss')
```

```
--- Experiment 2: Varying Neuron Count ---
Training model with 32 neurons per layer...
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dir
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 80ms/step - accuracy: 0.6416 - auc: 0.7359 - loss: 0.6245 - precision: 0.5990 - recall: 0.9152 -
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.8796 - auc: 0.9413 - loss: 0.3387 - precision: 0.8778 - recall: 0.8803 -
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9115 - auc: 0.9680 - loss: 0.2362 - precision: 0.9132 - recall: 0.9082 -
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9440 - auc: 0.9835 - loss: 0.1691 - precision: 0.9452 - recall: 0.9420 -
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.9537 - auc: 0.9886 - loss: 0.1385 - precision: 0.9571 - recall: 0.9494 -
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9692 - auc: 0.9930 - loss: 0.1051 - precision: 0.9712 - recall: 0.9668 -
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 56ms/step - accuracy: 0.9770 - auc: 0.9947 - loss: 0.0873 - precision: 0.9787 - recall: 0.9749 -
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - accuracy: 0.9768 - auc: 0.9956 - loss: 0.0781 - precision: 0.9768 - recall: 0.9764 -
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - accuracy: 0.9730 - auc: 0.9939 - loss: 0.0840 - precision: 0.9746 - recall: 0.9711 -
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9781 - auc: 0.9957 - loss: 0.0704 - precision: 0.9811 - recall: 0.9747 -
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 41ms/step - accuracy: 0.9871 - auc: 0.9970 - loss: 0.0498 - precision: 0.9867 - recall: 0.9875 -
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 41ms/step - accuracy: 0.9913 - auc: 0.9980 - loss: 0.0371 - precision: 0.9922 - recall: 0.9903 -
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9954 - auc: 0.9992 - loss: 0.0225 - precision: 0.9965 - recall: 0.9942 -
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 41ms/step - accuracy: 0.9967 - auc: 0.9991 - loss: 0.0191 - precision: 0.9972 - recall: 0.9962 -
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9989 - auc: 0.9996 - loss: 0.0115 - precision: 0.9995 - recall: 0.9983 -
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 56ms/step - accuracy: 0.9992 - auc: 0.9996 - loss: 0.0083 - precision: 1.0000 - recall: 0.9984 -
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 59ms/step - accuracy: 0.9807 - auc: 0.9952 - loss: 0.0712 - precision: 0.9843 - recall: 0.9768 -
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 41ms/step - accuracy: 0.9994 - auc: 1.0000 - loss: 0.0046 - precision: 1.0000 - recall: 0.9988 -
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9997 - auc: 1.0000 - loss: 0.0029 - precision: 1.0000 - recall: 0.9994 -
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 1s 40ms/step - accuracy: 0.9708 - auc: 0.9931 - loss: 0.1060 - precision: 0.9749 - recall: 0.9663 -
Training model with 64 neurons per layer...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 128ms/step - accuracy: 0.6735 - auc: 0.7286 - loss: 0.6096 - precision: 0.6646 - recall: 0.7137
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - accuracy: 0.8810 - auc: 0.9434 - loss: 0.3234 - precision: 0.8766 - recall: 0.8853 -
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - accuracy: 0.9117 - auc: 0.9689 - loss: 0.2324 - precision: 0.9123 - recall: 0.9099 -
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - accuracy: 0.9280 - auc: 0.9786 - loss: 0.1891 - precision: 0.9280 - recall: 0.9271 -
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 59ms/step - accuracy: 0.9501 - auc: 0.9881 - loss: 0.1406 - precision: 0.9507 - recall: 0.9489 -
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 0.9524 - auc: 0.9906 - loss: 0.1233 - precision: 0.9536 - recall: 0.9504 -
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 59ms/step - accuracy: 0.9679 - auc: 0.9939 - loss: 0.0966 - precision: 0.9709 - recall: 0.9644 -
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - accuracy: 0.9825 - auc: 0.9978 - loss: 0.0606 - precision: 0.9838 - recall: 0.9808 -
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 59ms/step - accuracy: 0.9820 - auc: 0.9967 - loss: 0.0617 - precision: 0.9845 - recall: 0.9793 -
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - accuracy: 0.9894 - auc: 0.9986 - loss: 0.0382 - precision: 0.9904 - recall: 0.9882 -
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - accuracy: 0.9959 - auc: 0.9996 - loss: 0.0220 - precision: 0.9973 - recall: 0.9944
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 58ms/step - accuracy: 0.9988 - auc: 0.9999 - loss: 0.0144 - precision: 0.9998 - recall: 0.9978 -
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - accuracy: 0.9991 - auc: 0.9999 - loss: 0.0084 - precision: 0.9998 - recall: 0.9983 -
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 57ms/step - accuracy: 0.9780 - auc: 0.9877 - loss: 0.1267 - precision: 0.9675 - recall: 0.9893 -
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - accuracy: 0.9997 - auc: 1.0000 - loss: 0.0033 - precision: 1.0000 - recall: 0.9995 -
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 62ms/step - accuracy: 0.9999 - auc: 1.0000 - loss: 0.0018 - precision: 1.0000 - recall: 0.9998 -
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 85ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 8.3543e-04 - precision: 1.0000 - recall: 1.00
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 63ms/step - accuracy: 0.9805 - auc: 0.9923 - loss: 0.0960 - precision: 0.9841 - recall: 0.9765 -
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 57ms/step - accuracy: 0.9999 - auc: 1.0000 - loss: 7.7006e-04 - precision: 1.0000 - recall: 0.99
```

```
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 2s 57ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 4.1006e-04 - precision: 1.0000 - recall: 1.00
Training model with 128 neurons per layer...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 123ms/step - accuracy: 0.6759 - auc: 0.7206 - loss: 0.6178 - precision: 0.6801 - recall: 0.6633
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 111ms/step - accuracy: 0.8609 - auc: 0.9297 - loss: 0.3407 - precision: 0.8636 - recall: 0.8557
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 102ms/step - accuracy: 0.9016 - auc: 0.9650 - loss: 0.2431 - precision: 0.8975 - recall: 0.9061
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 0.9256 - auc: 0.9788 - loss: 0.1878 - precision: 0.9276 - recall: 0.9225 -
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 0.9440 - auc: 0.9874 - loss: 0.1434 - precision: 0.9460 - recall: 0.9411 -
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 0.9696 - auc: 0.9944 - loss: 0.0928 - precision: 0.9716 - recall: 0.9670 -
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 87ms/step - accuracy: 0.9742 - auc: 0.9954 - loss: 0.0803 - precision: 0.9760 - recall: 0.9720 -
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 87ms/step - accuracy: 0.9829 - auc: 0.9959 - loss: 0.0610 - precision: 0.9810 - recall: 0.9848 -
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 125ms/step - accuracy: 0.9974 - auc: 0.9998 - loss: 0.0144 - precision: 0.9985 - recall: 0.9962
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 86ms/step - accuracy: 0.9900 - auc: 0.9968 - loss: 0.0385 - precision: 0.9915 - recall: 0.9884 -
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 87ms/step - accuracy: 0.9995 - auc: 1.0000 - loss: 0.0059 - precision: 0.9999 - recall: 0.9990 -
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 109ms/step - accuracy: 0.9997 - auc: 1.0000 - loss: 0.0021 - precision: 0.9999 - recall: 0.9995
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9957 - auc: 0.9970 - loss: 0.0331 - precision: 0.9974 - recall: 0.9939 -
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 86ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0034 - precision: 1.0000 - recall: 1.0000 -
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 87ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0010 - precision: 1.0000 - recall: 1.0000 -
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 132ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 4.4659e-04 - precision: 1.0000 - recall: 1.0
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 2.3514e-04 - precision: 1.0000 - recall: 1.00
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 1.3993e-04 - precision: 1.0000 - recall: 1.00
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 87ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 9.6189e-05 - precision: 1.0000 - recall: 1.00
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 3s 114ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 7.3147e-05 - precision: 1.0000 - recall: 1.0
Training model with 256 neurons per layer...
Epoch 1/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 8s 212ms/step - accuracy: 0.6680 - auc: 0.7016 - loss: 0.6493 - precision: 0.6830 - recall: 0.6141
Epoch 2/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 8s 143ms/step - accuracy: 0.8334 - auc: 0.9080 - loss: 0.3900 - precision: 0.8380 - recall: 0.8276
Epoch 3/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 191ms/step - accuracy: 0.8976 - auc: 0.9611 - loss: 0.2538 - precision: 0.8907 - recall: 0.9055
Epoch 4/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 145ms/step - accuracy: 0.9255 - auc: 0.9795 - loss: 0.1848 - precision: 0.9260 - recall: 0.9240
Epoch 5/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 4s 146ms/step - accuracy: 0.9451 - auc: 0.9868 - loss: 0.1442 - precision: 0.9449 - recall: 0.9446
Epoch 6/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 165ms/step - accuracy: 0.9694 - auc: 0.9940 - loss: 0.0918 - precision: 0.9694 - recall: 0.9690
Epoch 7/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 155ms/step - accuracy: 0.9806 - auc: 0.9980 - loss: 0.0564 - precision: 0.9819 - recall: 0.9792
Epoch 8/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 191ms/step - accuracy: 0.9949 - auc: 0.9998 - loss: 0.0307 - precision: 0.9985 - recall: 0.9911
Epoch 9/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 9s 163ms/step - accuracy: 0.9972 - auc: 0.9999 - loss: 0.0135 - precision: 0.9978 - recall: 0.9967
Epoch 10/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 170ms/step - accuracy: 0.9610 - auc: 0.9733 - loss: 0.2278 - precision: 0.9710 - recall: 0.9497
Epoch 11/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 178ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0022 - precision: 1.0000 - recall: 1.0000
Epoch 12/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 10s 153ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 6.2139e-04 - precision: 1.0000 - recall: 1.
Epoch 13/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 7s 219ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 2.5086e-04 - precision: 1.0000 - recall: 1.0
Epoch 14/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 8s 154ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 1.3032e-04 - precision: 1.0000 - recall: 1.0
Epoch 15/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 189ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 8.4781e-05 - precision: 1.0000 - recall: 1.0
Epoch 16/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 11s 213ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 6.3047e-05 - precision: 1.0000 - recall: 1.
Epoch 17/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 8s 144ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 5.0251e-05 - precision: 1.0000 - recall: 1.0
Epoch 18/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 6s 174ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 4.1750e-05 - precision: 1.0000 - recall: 1.0
Epoch 19/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 10s 180ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 3.5765e-05 - precision: 1.0000 - recall: 1.
Epoch 20/20
30/30 ━━━━━━━━━━━━━━━━━━━━ 5s 152ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 3.1238e-05 - precision: 1.0000 - recall: 1.0
```

Neuron Count - Accuracy

```
# Experiment 3: Comparing loss functions
print("\n--- Experiment 3: Loss Function Comparison ---")
histories_loss_functions = {}
print("Training model with MSE loss function...")
mse_model = construct_model(loss_fn='mse')
history_mse = mse_model.fit(x_partial_train, y_partial_train,
epochs=20,
batch_size=512,
validation_data=(x_validation, y_validation),
verbose=1)
histories_loss_functions['MSE'] = history_mse
val_acc = history.history['val_accuracy'][-1]
test_acc = model.evaluate(x_test, y_test, verbose=0)[1]
experiment_results['Neuron Count'][neurons] = {'Validation Accuracy': val_acc, 'Test Accuracy': test_acc}

visualize_metrics(histories_loss_functions, 'Loss Function Comparison - Accuracy', 'Accuracy')
visualize_metrics(histories_loss_functions, 'Loss Function Comparison - Loss', 'Loss')
```